

SECTION D: PREDICTIVE MODEL DEVELOPMENT & UNCERTAINTY QUANTIFICATION

MEMORANDUM

- **TO:** MSAC512 Assessment Board & Lead Actuarial Examiner
- **FROM:** Trainee Data Analyst
- **DATE:** 23 September 2026
- **SUBJECT:** Predictive Model Development, Spatial Stratification, and Uncertainty Quantification

5.1 Actuarial Benchmark Model (GLM Baseline)

To establish an actuarial ratemaking baseline, a Generalized Linear Model (Log-OLS) was fitted on the training split ($n = 171$) using only structured physical and spatial fields: strat_suburb, log_stand_size, bedrooms, and bathrooms. Suburbs with fewer than 5 observations (e.g., Mabelreign, Vainona) were combined into an 'Other' reference tier to enforce stratification stability.

$$\ln(E[\text{Price}_i]) = \beta_0 + \beta_1 \ln(\text{Stand Size}_i) + \beta_2 \text{Bedrooms}_i + \beta_3 \text{Bathrooms}_i + \sum_k \gamma_k \mathbb{I}(\text{Suburb}_i = k)$$

Full Model Coefficients & Direct Responses ($e^\beta - 1$):

Variable / Parameter	Fitted Coef (β)	Std Error	t-statistic	p-value ($P> t $)	Multiplicative Impact ($e^\beta-1$)	Actuarial Interpretation
Intercept (β_0)	4.6580	0.224	20.808	< 0.001	—	Baseline log valuation anchor.
log_stand_size	+0.4945	0.051	9.609	< 0.001	+63.96%	A 100% increase in land area increases

Variable / Parameter	Fitted Coef (β)	Std Error	t-statistic	p-value ($P> t $)	Multiplicative Impact ($e^{\beta}-1$)	Actuarial Interpretation
						property value by +49.45%.
bedrooms	-0.1904	0.102	-1.863	0.064	-17.34%	Each additional bedroom holding land constant reduces value by -17.34% (indicating land overcrowding) .
bathrooms	+4.6580	0.224	20.808	< 0.001	+10,442.2%	Large multiplicative factor associated with modern en-suite developments.
Suburb: Borrowdale	+0.0075	0.226	0.033	0.973	+0.75%	Indistinguishable from baseline suburb in raw structured form.
Suburb: Chitungwiza	-1.0624	0.575	-1.848	0.066	-65.43%	High-density outlying location discount.

Variable / Parameter	Fitted Coef (β)	Std Error	t-statistic	p-value ($P> t $)	Multiplicative Impact ($e\beta-1$)	Actuarial Interpretation
Suburb: Harare	-0.1349	0.342	-0.394	0.694	-12.62%	Central city baseline.
Suburb: Mabvazuva	-0.4650	0.308	-1.508	0.134	-37.19%	High-density expansion discount.
Suburb: Madokero	-0.2598	0.350	-0.742	0.459	-22.88%	High-density planned estate discount.
Suburb: Mount Pleasant	+0.3479	0.569	0.611	0.542	+41.61%	Established upmarket premium.
Suburb: Norton	-0.8865	0.402	-2.205	0.029	-58.79%	Outlying satellite town discount.
Suburb: Ruwa	-0.5020	0.369	-1.362	0.175	-39.47%	Peripheral suburban discount.
Suburb: Other	+0.0238	0.346	0.069	0.945	+2.41%	Grouped sparse suburb baseline.

5.2 Multimodal Machine Learning Suite

Three multimodal models incorporating unstructured textual features (TF-IDF n -grams and extracted flags) and visual completion/tiling scores were fitted alongside structured fields:

1. **ElasticNet Regularized Linear Model:** Applies L_1/L_2 penalties (α tuned via 5-fold CV) to handle collinearity across engineered features.

2. **Random Forest Regressor:** An ensemble of 300 non-parametric decision trees capturing non-linear feature interactions without linear functional assumptions.
3. **XGBoost Regressor:** A gradient-boosted decision tree architecture ($\text{max_depth} = 4, \eta = 0.03$) sequentially fitting residual errors.

5.3 Stratified Validation Protocol & Actuarial Metrics

A. Suburb-Stratified Split Execution

To prevent spatial data leakage, properties were split using suburb stratification (70% Train [$n = 171$], 15% Validation [$n = 37$], 15% Test [$n = 37$]) after grouping sparse suburbs ($n < 5$) into an 'Other' category.

B. Actuarial Metrics

Models were evaluated on the original USD scale ($\hat{y} = \exp(\ln(\hat{y}))$) using:

- **RMSE (Root Mean Squared Error):** Penalizes extreme pricing outliers.
- **MAE (Mean Absolute Error):** Average absolute error in USD.
- **MAPE (Mean Absolute Percentage Error):** Relative error across lower- and higher-value properties.

5.4 Prediction Intervals & Uncertainty Quantification

Using **Split Conformal Prediction** calibrated on the validation set ($n = 37$), non-conformity residuals $s_i = |y_i - \hat{y}_i|$ at the 90th percentile ($q_{0.90}$) were computed to construct 90% prediction intervals on unseen test data:

$$\hat{C}(X_{\text{test}}) = [\max(0, \hat{y}_{\text{test}} - q_{0.90}), \hat{y}_{\text{test}} + q_{0.90}]$$

5.5 Final Model Comparison & Trade-Off Analysis

Below is the model performance comparison across the holdout test set ($n = 37$):

Model Architecture	Holdout RMSE (USD)	Holdout MAE (USD)	Holdout MAPE (%)	R2 Score	90% Prediction Interval Width (USD)	Empirical Test Coverage (%)
GLM Benchmark (Log-OLS)	\$928,986.81	\$430,880.35	418.55%	−0.854	\$1,661,216.29	91.9%
ElasticNet Regularized	\$799,083.74	\$365,419.75	278.70%	−0.371	\$1,077,032.77	86.5%
Random Forest Regressor	\$694,396.78	\$307,660.36	252.81%	-0.036	\$2,016,483.77	94.6%
XGBoost Regressor	\$700,818.80	\$321,157.14	252.55%	−0.055	\$2,105,611.80	94.6%

Justification & Model Selection Rationale

1. Failure of Structured GLM Baseline:

The structured GLM exhibits severe underfitting ($R^2 = -0.854$, $\text{MAPE} = 418.55\%$), heavily mispricing multi-million dollar luxury estates because it lacks visual and textual finish condition inputs.

2. Multimodal Machine Learning Gains:

Incorporating image quality scores and text flags reduces test MAE by over \$123,000 (from \$430,880.35 down to \$307,660.36) and reduces relative error (MAPE) from 418.55% to 252.81%.

3. Final Model Selection — Random Forest Regressor:

Random Forest is selected as the optimal model. It achieves the lowest overall RMSE (\$694,396.78), lowest MAE (\$307,660.36), and superior R^2 performance (-0.036) compared to XGBoost (-0.055) and ElasticNet (-0.371), while providing 94.6% empirical interval coverage.